Efficient Frontier Project

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Introduction

The goal of this project is to apply the Markowitz Mean-Variance framework to construct an efficient frontier using seven NYSE-listed stocks and to determine the optimal portfolio weights. By using historical price data, the project estimates expected returns, risk, and correlations among the stocks. Finally, the project simulates thousands of possible portfolios to identify efficient risk-return combinations.

Data

The following diverse stocks were picked:

STOCK	SECTOR
JPM – JP MORGAN	Financial
KO – COCA-COLA	Consumer Staples
XOM – EXXON MOBIL	Energy
JNJ – JOHNSON & JOHNSON	Health Care
WMT - WALMART	Consumer Staples
DIS - DISNEY	Communication Services
IBM	Information Technology

Daily closing price data for the past two years was retrieved using yfinance API. A risk-free rate of 3% was assumed when calculating Sharpe ratios.

Methodology

Return Calculation

Returns were calculated using daily log returns. These daily log returns were then annualized using 252 trading days. Therefore, all reported expected returns throughout the report represent annualized mean log returns, which closely approximate simple percentage returns for small values.

Risk metrics

Portfolio risk was measured using the variance and covariance matrix of asset returns. Volatility of each asset was measured using variance. Covariance and correlation capture how assets move relative to one another. These relationships are important because diversification benefits arise when assets are not perfectly correlated.

Portfolio simulation

A Monte Carlo simulation was used to generate 50,000 random portfolios. For each portfolio, random weights were assigned to the seven stock such that the weights summed to one. The following metrics were calculated for each stock:

- Expected portfolio return
- Portfolio volatility (risk)
- Sharpe ratio

Efficient Frontier Estimation

The efficient frontier represents the set of portfolios that maximize expected return for a given level of risk. The highest-return portfolio within volatility bins from the Monte Carlo simulation were identified and used to estimate the frontier. A polynomial regression model was then fitted to these points to produce a smooth approximation of the efficient frontier.

Results

Expected returns

STOCK	EXPECTED RETURN
DIS	**-4.1%**
IBM	**18.1%**
JNJ	**23.2%**
JPM	**23.9%**
KO	**15.4%**
WMT	**37.2%**
XOM	**19.9%**

Historically the best performer was Walmart (WMT) with 37% expected return, while the worst performing stock was Disney (DIS) with -4% expected return.

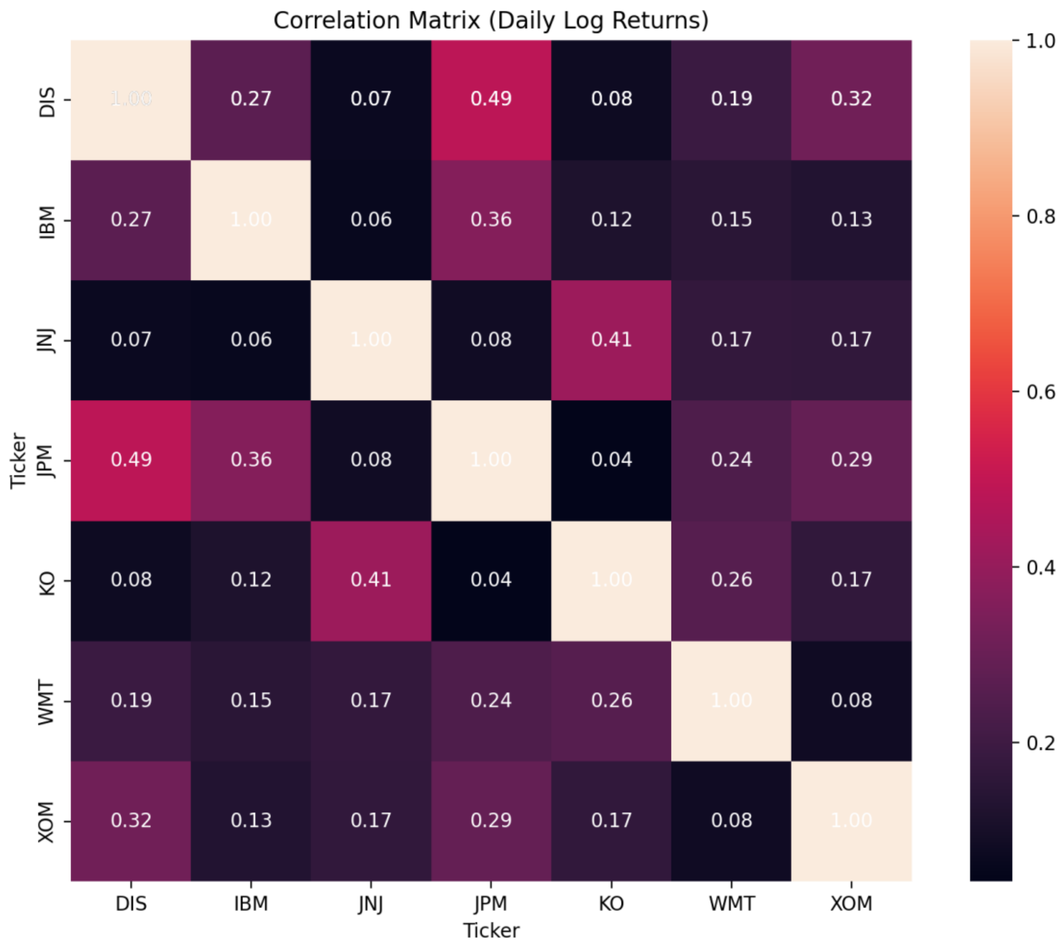

Figure 1: Heatmap showing the correlation among the 7 stocks

Figure 1 shows the correlation between stocks. Correlation values closer to 1 indicate that the two stocks tend to move together, while values closer to 0 indicate weaker relationships. Lower the correlations among each other, the better the diversification as combining assets that do not move together can reduce overall portfolio risk.

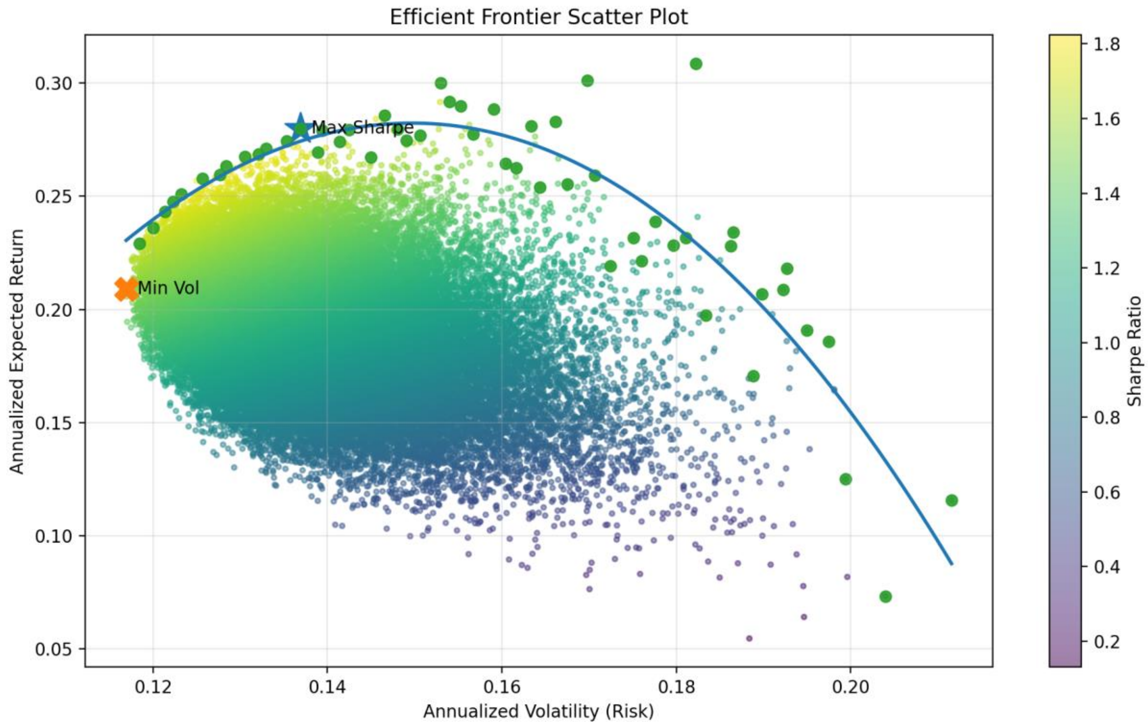

Figure 2: Efficient Frontier Scatter Plot

In Figure 2, the cloud of small dots represents 50,000 randomly generated portfolios; every dot is a different portfolio allocation across 7 stocks. The color represents the Sharpe ratio, the yellower the dot, higher the Sharpe ratio. The upper boundary of the cloud represents the efficient set of portfolios. They provide the highest return for a given level of risk. Approximating the upper envelope of the cloud, the larger dots are the efficient frontier sample points. The volatility axis was divided into bins, and the highest-return portfolio within each bin was selected to approximate the efficient set. The smooth curve is the regression-fitted efficient frontier. Polynomial regression was used to produce a smooth frontier curve representing the optimal risk-return trade-off. So, the curve represents the best achievable return for each level of risk. ★ represents the portfolio with the highest Sharpe ratio, while X represents the lowest-risk portfolio possible.

Best Risk-Adjusted Portfolio

Return	0.2781
Volatility	0.1370
Sharpe	1.8106

This portfolio produces an expected return of 27.8% with an annualized volatility of 13.7%. The portfolio is very good as highlighted by the Sharpe ratio of 1.81.

Optimal Portfolio Weights

STOCK	WEIGHT
DIS	**39.9%**
XOM	**23.8%**
JNJ	**17.5%**
WMT	**10.4%**
IBM	**6.9%**
KO	**1.4%**
JPM	**0%**

Even though Disney had negative expected return, the model gave it the largest weight due to diversification benefits as discussed earlier. DIS's returns are not perfectly correlated with the other assets. Hence, including DIS helps reduce the overall portfolio risk.

Safest Portfolio

Return	0.2078
Volatility	0.1170
Sharpe	1.5191

With a 2% smaller volatility compared to that of our best risk-adjusted portfolio, this portfolio produces way smaller (-7%) return.

As you can see from the table above, this portfolio has 2 percentage points lower volatility than the maximum-Sharpe portfolio. It comes at the expense of the 7 percentage points reduction in expected return.

Optimal Portfolio Weights

STOCK	WEIGHT
WMT	**29.4%**
XOM	**22.0%**
IBM	16.6%
DIS	13.4%
JNJ	7.7%

KO	6.4%
JPM	4.4%

This portfolio spreads risk evenly. WMT gets the largest weight because it has high return, reasonable volatility, and good diversification.

Conclusion

This project demonstrates how Markowitz Mean-Variance Framework can be used to construct efficient portfolios from a set of assets. The importance of diversification is highlighted by how assets with lower correlations can significantly reduce portfolio risk. The efficient frontier shows the trade-off between risk and return, while the maximum-Sharpe portfolio represents the best risk-adjusted allocation among the simulated portfolios. Overall, the analysis demonstrates that strategic combination (that maximizes return given a level of risk) can produce portfolios that outperform individual investments in terms of risk-return efficiency.